

Experiment NO : 09
Experiment Name : Study of Differentiator and Integrator circuit using OP-Amp

Objectives:

- To implement the Differentiator and Integrator circuit using Op-Amp
- To observe the input and output waveforms of the differentiator and integrator circuit

Theory:

□ Differentiator Circuit:

A Differentiator Circuit produces an output voltage that is proportional to the rate of change of the input voltage. The basic Op-Amp differentiator is implemented using a Capacitor in series with the input and a resistor in the feedback path

□ Integrator Circuit:

An Integrator Circuit produces an output voltage that is proportional to the time integral of the input voltage. The basic Op-Amp integrator is implemented using a resistor in series with the input and a capacitor in the feedback path

The input/output relationship of DC and IC are provided by,

$$V_o = -R_f C \frac{dV_i}{dt} \quad \text{and} \quad V_o = -\frac{1}{RC} \int V_i dt$$

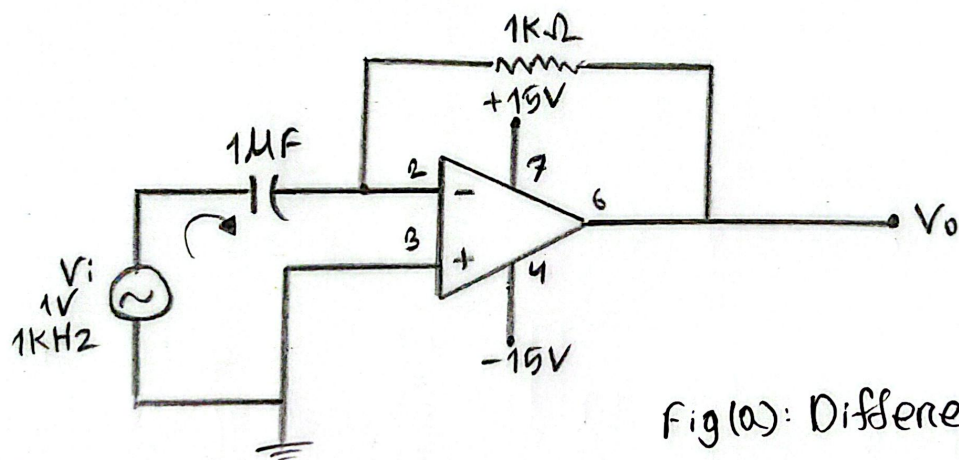
Differentiator

Integrator

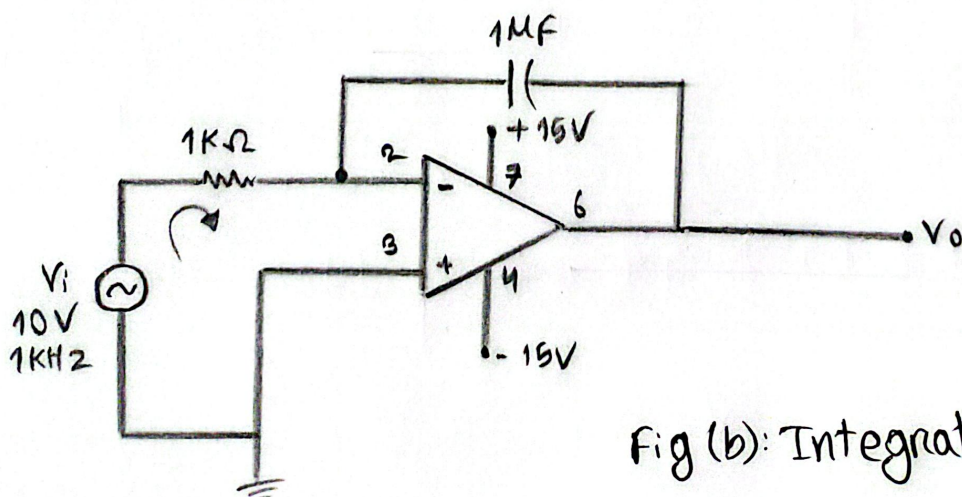
Required Apparatus:

- Op-Amp (u741, 1 piece)
- Resistor ($1k\Omega$: 1 piece)
- Capacitor ($1\mu F$: 1 piece)
- Signal Generator
- Oscilloscope
- DC Power supply (0-15V)
- Project Board
- Multimeter
- Connecting wires

Circuit Diagram:



Fig(a): Differentiator Circuit



Fig(b): Integrator Circuit

Procedure:

- (i) The circuit was connected according to Fig(a)
- (ii) An AC signal was supplied from the signal generator and the corresponding output waveform was recorded
- (iii) Again the circuit was connected according to Fig(b) and step (ii) was repeated

Calculations:

Differentiator Circuit:

$$\text{Theoretically, } A_v = 2\pi = 6.283$$

$$\begin{aligned}\text{Practically, } A_v &= \frac{V_o}{V_i} \\ &= \frac{11.84}{1.84} \\ &= 6.435\end{aligned}$$

$$\begin{aligned}\text{Error} &= \frac{6.435 - 6.283}{6.283} \times 100\% \\ &= 2.42\%\end{aligned}$$

Integrator Circuit:

$$\text{Theoretically, } A_v = \frac{1}{2\pi} = 0.159$$

$$\begin{aligned}\text{Practically, } A_v &= \frac{V_o}{V_i} \\ &= \frac{3.2}{20} \\ &= 0.16\end{aligned}$$

$$\begin{aligned}\text{Error} &= \frac{0.16 - 0.159}{0.16} \times 100\% \\ &= 0.62\%\end{aligned}$$

Discussion and Conclusion:

The Differentiator and Integrator circuits operated as expected. The small errors (2.42% and 0.62%) confirms good arrangement between theory and experiment

Precautions:

- All connections were given properly
- OP-Amp terminals were ~~be~~ identified properly
- Calibration was performed correctly